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Thermodynamic analysis of biomedical waste plasma gasification

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ABSTRACT

Plasma gasification technology is one of the environmentally correct techniques that can be applied in the processing of biomedical waste (BW). This work aims to present thermodynamic studies with a simulation of the plasma gasification of BW produced in Brazil. Through thermodynamic analysis is determined the best operating point of the reactor, which corresponds to the temperature where the energy yield of syngas is maximum, and consequently the syngas chemical composition and its lower heating value (LHV). Finally, it is estimated the electrical power required in the BW processing and the potential for electricity generation through the burning of syngas in an internal combustion engine (ICE) and gas turbine set (GTS), and the capacity to supply the necessary energy in the plasma gasifier. As conclusion, the best operating point for the processing of typical Brazilian BW is at a temperature of 1040 K with a maximum gas energy yield of 2.25. For this temperature the syngas consists of 63.65 wt% of carbon monoxide and 5.35 wt% of hydrogen and LHV of 13.47 MJ/kg. Finally, for processing 1 kg/s of BW are required 6292 kW of electrical power, and the maximum electricity production potential is 3132 kW in ICE and 3758 kW in GTS.

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1. Introduction

The Covid-19 pandemic devastated the world and brought great concern to public health. An increase in the number of cases of infected people and the hospitalization requirement determined a quick construction of field hospitals. Analyzing of biomedical waste generation in Brazil, Agência Brasil [1] shows that there was an increase in 20% in the June 2020 compared to the same period of the previous year before the pandemic. Likewise, the municipal government of the city of São Paulo [l] (SP, Brazil), pointed to a 10% growth in the generation of biomedical waste (RB) in 2020 compared to 2019, and indicated a new growth in the year 2021 [2]. The inadequate management of BW together with the pandemic poses a serious threat to the health of the population and the environment for the present and future.

Biomedical waste is the waste originating from the activities of health service units, providers of medical, dental, laboratory, pharmaceutical including educational and research institutions linked to both the human and veterinary. The BW consists of expired vaccines, blood and blood products, tissues, sharp materials, body fluids, radioactive waste, chemical and pharmaceutical waste, and protective masks which increased during the pandemic, [3].

Cafure and Patriarcha-Gracioli [4], affirm that the environmental impacts caused by the inadequate management of contaminated BW can increase hospital infection rates or provoke epidemics as a result of the groundwater contamination. Thus, it is essential that technologies aimed at ecologically correct and effective disposal and treatment (transformation into inert waste) of BW are developed and applied in all municipalities.

According to the latest Panorama of Solid Waste in Brazil 2018/2019 [5], among all 5,570 Brazilian municipalities [6], only 81.5% collected BW in 2018. Furthermore, only 63.8% of the collected BW had an adequate destination, the others were destined without prior treatment to dumps, landfills, septic ditches, etc. In numbers, in 2018, it was collected 252,948 tons of BW, which is equivalent to

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Nomenclature

$\dot{E}_p$:	electrical power [kW]
$\dot{E}_{gas}$	energy supplied by the gas [kW]
E_{torch}	electrical power required in the torch [kW]
LHV_{gas}	gas lower heating value [kJ/kg]
$\dot{m}_{ashes} = \dot{m}_3$	fluxo mássico de cinzas [kg/s]
$\dot{m}_{BW} = \dot{m}_1$	fluxo mássico de RSS [kg/s]
$\dot{m}_{gas}$	gas mass flow [kg/s]
$\dot{m}_{other\ gases}$	other gases mass flow [kg/s]
$\dot{m}_{oxy}$	oxidant mass flow [kg/s]
$\dot{m}_{syngas}$	syngas mass flow [kg/s]
$\dot{m}_W$	organic part mass flow [kg/s]
X_i	the mass fractions of i-th species of the mixture [−]
α :	energy yield of gas [−]
η_{el}	electricity generation efficiency [−]
η_{torch}	torch efficiency [−]
ΔH	variation of the enthalpy [kJ/kg]

1.2 kg per inhabitant per year [5].

Fig. 1 shows the BW final destination collected by Brazilian municipalities.

Fig. 1 shows the technologies applied in the BW treatment collected in Brazilian municipalities. It can be seen that the main destinations are incineration (40.2%) and "others" (36.2%), which represents a final destination in dumps, landfills, septic ditches, etc.

1.1. Conventional technologies applied in biomedical waste treatment

The incineration has been the most common method for treating BW. The advantage of incineration is the decontamination of waste and a significant reduction in the volume destined for the landfill, which reaches about 10% of the original value [7]. However, some types of biomedical waste are difficult to incinerate (for example, full urine bags and body parts) and need special attention. Moreover, the incineration of solid healthcare waste is one of the main sources of pollution by dioxins and furans, in particular due to the presence of PVC products [8,9].

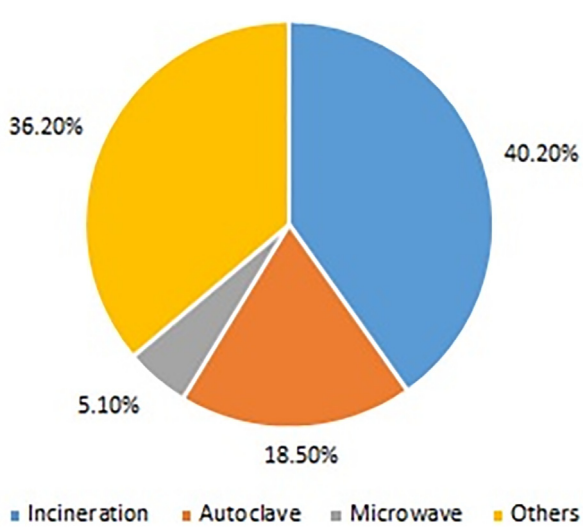

Fig. 1. Distribution of technologies applied in the BW treatment in Brazil (adapted [5]).

The autoclaving and microwave system are fully suitable disinfection technologies for the treatment of BW, but there is no reduction in the volume of waste, so the crunch process is normally applied to reduce the volume by 60−80% before landfill [7]. These technologies only disinfect the BW, and do not take advantage of an energy power of BW.

Another promising technology that can be applied to BW processing is gasification (fluidized bed, fixed bed), which converts waste into fuel gas, through chemical reactions in a reducing environment with different gasification agents (air, oxygen, water steam, carbon dioxide, etc). In the case of fluidized bed gasifiers, for example, a relatively low temperature is used to prevent agglomeration and sintering of bed ash during processing. Thus, the gas produced by this kind of gasifiers has in its composition tars and other condensable organic species that create obstacles in the process and require costly and complex techniques to be removed. Furthermore, the produced ash can contain high levels of fixed carbon, heavy metals and organic pollutants that reduce the process's conversion efficiency and limit its any secondary use [10]. These disadvantages are a limiting factor for the use of this technology in biomedical waste processing.

In view of the disadvantages of the technologies mentioned above together with inadequate disposal of biomedical waste puts on the agenda the demand for the development and dissemination of technologies that eliminate or at least reduce discard volumes and recover the energy contained in BW. According to Ricardo Ricchini from the Setor Reciclagem website [11], the biggest problem of BW is the infective waste of class A and B, which represents a high risk of contamination, in addition to the environment polluting. Thus, new technologies need to be studied and disseminated so that the BW can be completely destroyed and still use its energy, such as the plasma gasification technology.

1.2. Plasma gasification technology

Plasma gasification seems to be the most suitable technology for BW processing, as it has several advantages, including: a high thermal energy density (which leads to smaller reactor dimensions and results in lower thermal losses), easy control of reactor temperature, the possibility to operate independently of the environment gases (ie air, inert gases, water steam, etc), reduction emission of polluting gases and treated BW volume [12].

The plasma is a source of the energy necessary for BW gasification. Due to the high temperature of the process, plasma gasification ensures a fast and efficient deactivation of pathogenic microorganisms and conversion of organic material, common in hospital waste [12]. The organic components (mainly carbon, hydrogen and oxygen) are volatilized and complex molecules dissociated, producing a mixture composed mainly of CO and H_2 (well known as syngas) with tiny amounts of pollutant gases. This gas (syngas) can be used as high quality fuel [13].

Messerle [9] presents in his work a simulation of the gasification process considering a chemical composition of bony tissue (BT) and household waste (HW), as shown in Table 1, using air and steam as oxidant. Under these conditions, the chemical composition of the gas produced and the lower heating value is obtained according to

Table 1
Chemical composition of HW and BT, wt.% [9].

Waste	C	H	O	N	S	Cl
BT	9.00	2.21	1.99	4.00	1.00	–
HW	34.15	5.85	6.29	8.16	0.94	5.3

the temperature and the maximum production of CO and H_2, according to Table 2.

In the plasma gasification process, in addition to producing syngas, it also produces ash (slag), which is collected at the bottom of the reactor continuously or periodically [11]. The plasma enables attain the temperatures above the ash melting point and inertizar it by vitrification, [9]. In this case, the removal can be done in molds to create ingots or bricks (for use in pavements) or tempered in water to create granules to be reused or discarded in landfills.

The plasma gasification system consists of the following components: a pre-treatment and feed system of raw material, the plasma torche(s) coupled with energy sources, a gasification reactor, a cooling system for plasma torches and the system components subjected to elevated temperatures, a removal system, a produced gas treatment system and a monitoring and control system. The entire process must be monitored and controlled to ensure optimal performance of reactor temperature, plasma power input, raw material feed rate and ash removal [14].

Plasma gasification reactors have various configurations, but generally consist of a reactor chamber connected to one or several plasma torches that supply the energy needed for the gasification reactions. The inner wall of the reactor chamber is covered with refractory material capable of withstanding high temperatures from the plasma jet or electric arc.

According to Minutillo [15], before being used as fuel in an ICE or GT cycle, the syngas must be cooled and cleaned. Thus, the cycle combined with both ICE and GT must have a cooling and cleaning system right after the plasma gasifier.

Given the need that Brazil and the world have to find a more attractive BW treatment or destruction option, such as the plasma gasification technology, this work aims to show studies of plasma gasification of Brazilian BW. Through the TERRA software, the plasma gasification process is simulated considering the typical chemical composition of Brazilian BW. Through this is obtained the chemical composition of syngas to the temperature of 300−2300 K and by calculating the energy yield, which is a ratio between the chemical energy contained in the gas and the energy spent in the process, the best operating point is determined of the plasma gasification of Brazilian BW. This information that Brazil does not have yet and that there are few studies related to the BW gasification process in other countries. It is still the objective of this work is to investigate the potential of electricity production through gas burning (syngas) in an internal combustion engine and gas turbine set, so that previous studies are presented in relation to the capacity of the energy produced to meet the need for a plasma gasifier.

2. Materials and methods

2.1. Material

To perform the calculations it is necessary to know the composition of the raw material to be processed. In Brazil, there are few publications on the BW composition, due to its heterogeneity and its pathogenic character that presents a high degree of difficulty and limits the development of new research. In this work, two studies of BW characterization were considered. One was performed in a hospital (Hos) in the city of São Carlos, state of São Paulo [16] and the other in a hospital in the city of Florianópolis, state of Santa Catarina [17]. In the latter, the waste from different hospital centers was considered: the nursery; the surgery center and the Others, which include the emergency room, laboratories and administration center. Elemental (CHON) and immediate (moisture, ash) analysis of biomedical waste is presented in Table 3 together with averaged values and standard error.

Table 2
Concentration of gaseous, vol% [9].

Temperature	BT (air)	BT (steam)	HW (air)	HW (steam)
	1300 K		1600 K	
CO	28.7	38.1	31.7	33.6
H_2	24.7	46.8	50.9	60.8
CH_4	0.01	0.2	–	–
CO_2	2.46	0.25	0.13	0.13
H_2O	3.72	0.25	0.13	0.13
LHV [kJ/kg]	3510	5664	13,620	18,497

2.2. Methodology

2.2.1. Thermodynamic analysis

To carry out the technical feasibility study, a thermodynamic analysis of BW treatment is performed. Firstly, a chemical composition of the gas phase inside the plasma chemical reactor was evaluated on the assumption of thermodynamic equilibrium state. Then, the best operating temperature of the gasifier, corresponding to the highest energy yield of the gas, was determined. At this point, applying the mass conservation equation to the control volume formed by the plasma gasifier, the gas flow rate and composition of the produced were determined. These parameters serve as the input data to estimate the potential of electricity production in ICE and TG.

2.2.1.1. Gas production. In order to obtain the composition of the gas produced in the plasma reactor the state of thermodynamic equilibrium is assumed. There are two methods most used in calculation of the composition of a system in thermodynamic equilibrium. The first method is based on the law of mass conservation and uses equilibrium constants. For thermodynamic calculations, it doesn't matter the path taken by the process, only the initial and final states of the system are important. In this case, the chemical transformations are presented in the form of reversible reactions of dissociation of substances into primary elements and those of synthesis from these elements.

The second method is based on the search a state of equilibrium in which the concentrations of the various species no longer change spontaneously. In this state the system attains a maximum entropy value or minimum value of Gibbs free energy. In this case, a thermodynamic system is considered as a set of individual substances without considering chemical reactions. The extreme values of the characteristic functions (i.e. the entropy or the Gibbs free energy) are determined for all phases, considering the mass conservation laws, energy conservation (in the case of plasma) and the equation of state. To find the extremum of characteristic function and obtain the composition of the system is applied the Lagrange indefinite multipliers method. Having compiled the Lagrange function and taking the partial derivatives with respect to all variables, a system of equations is obtained and that can be solved by successive approximations. To apply this method is used the simulation of the plasma gasification of Brazilian BW presented in Table 3, considering pyrolysis, in the TERRA software. Thus is obtained the chemical composition of the syngas considering temperatures between 300 and 2300 K and atmospheric pressure of 0.1 MPa.

To determine the best operating temperature, from the point of view of energy production, the energy yield of gas was evaluated. The energy yield, α, is a ratio between the chemical energy contained in the gas (calculated by its Lower Heating Value, LHV_{gas}) and energy spent in the process, evaluated by the variation of the enthalpy ΔH of the thermodynamic system, according to Eq. (1)

Table 3
Biomedical waste composition from Brazilian (wt%).

	Hospital [16]	Infirmary [17]	Surgery Center [17]	Others [17]	Average	Std. Error
C	35.73	30.80	27.90	32.00	31.6	1.6
H	3.87	3.60	3.90	3.60	3.74	0.08
S	0.00	0.00	0.50	0.30	0.20	0.12
Cl	0.32	0.08	0.09	0.09	0.15	0.06
N	0.51	0.50	0.38	0.64	0.51	0.05
O	37.40	36.72	28.13	47.07	37.3	3.9
H_2O	18.33	24.10	28.60	12.20	20.8	3.6
Ashes	9.03	4.20	10.50	4.10	7.0	1.7

$$\alpha = \frac{LHV_{gas}}{\Delta H} \tag{1}$$

The lower heating value of the gas for each temperature was determined from the respective composition of the gas, considering the lower calorific value according to Eq. (2):

$$LHV_{gas} = \sum_i LHV_i X_i \tag{2}$$

Here LHV_i is the lower heating value and X_i is the mass fractions of i-th species of the mixture, respectively. The principal species of produced gas are hydrogen H_2 and carbon monoxide CO with LHV values of 120 and 11 MJ/kg, respectively [18].

2.2.1.2. Mass conservation. The quantities of processes BW and produced gas were determined from the mass conservation $\frac{dM}{dt} = 0$ applied to control volume represented by the inner volume of the plasma gasifier. Thus, the entrance mass flow consists of the BW mass flow rate $\dot{m}_{BW}$ and the mass flow of oxidizing gas $\dot{m}_{oxy}$ or the plasma forming gas, needed for the electric arc stabilization. On the output there are the mass flow of produced gases $\dot{m}_{gases}$ and the mass flow of ashes $\dot{m}_{ashes}$. The conservation of mass for the plasma gasifier is shown in Eq. (3)

$$\dot{m}_{BW} + \dot{m}_{oxy} = \dot{m}_{gas} + \dot{m}_{ashes} \tag{3}$$

The BW mass flow rate composed of the organic part $\dot{m}_W$ and ashes $\dot{m}_{ashes}$:

$$\dot{m}_{BW} = \dot{m}_W + \dot{m}_{ashes} \tag{4}$$

The last right hand term includes the BW inorganic part as well as the condensed species formed during plasma treatment. Its quantity depends on the processing temperature. The principal component of the produced gases is the syngas (H_2 and CO mixture).

$$\dot{m}_{gas} = \dot{m}_{syngas} + \dot{m}_{other\ gases} \tag{5}$$

Other gases include non combustible gases (like CO_2, H_2O, N_2, etc) and volatilized species.

2.2.1.3. Electric power required by the plasma torch. The electric power required by the plasma torch operated in gasifier can be determined as:

$$E_{torch} = \frac{\dot{m}_{BW}\ \Delta H}{\eta_{torch}} \tag{6}$$

The plasma torch efficiency, η_{torch}, to transform electrical energy in thermal one depends on the design and type of plasma torch and varied from 70% for the non transferred arc plasma torch to 95% for plasma torch with transferred arc. The plasma torches with non transferred arc require a relatively high plasma forming gas flow rate for stabilizing the electric arc. This kind of plasma torches is adequate to provide oxidant in the gasifier. On the contrary, the transferred arc plasma torches require much less flow rates of gas and are suitable for pyrolysis. Thus, in this work the transferred arc plasma torch was chosen.

2.2.1.4. Electricity production potential in ICE and GT. The plasma gasification process requires a electrical power, part of which can be recovered by burning the produced gas in ICE or TG. The electricity production in ICE and TG, $\dot{E}_p$, is estimated through the Eq. (7).

$$\dot{E}_p = \eta_{el}\ \dot{E}_{gas} \tag{7}$$

The energy supplied by the gas is determined through Eq. (8):

$$\dot{E}_{gas} = \dot{m}_{gas}\ LHV_{gas} \tag{8}$$

For ICE, the electricity generation efficiency (η_{el}) varied from 14% to 26%, while d for TG - from 20% to 30% [19]. For this work was considered the minimum and maximum value of the electricity generation efficiency for each equipment, in order to estimate the minimum and maximum potential of the recovered electrical power from produced gas.

3. Results

3.1. Thermodynamic analysis results

The thermodynamic analysis for the system in equilibrium state was carried out in the temperature range of 300–2300 K and atmospheric pressure of 0.1 MPa to determine the quantity and composition of the gas produced in the plasma gasification process. Firstly, the composition of the feedstock was analyzed. As follows from Table 3, the main components of BW are oxygen (37.3 wt%), carbon (31.6 wt%) and water (20.8 wt%).

For determination of the amount and composition of produced gas the thermodynamic analysis for the system in equilibrium state was carried out in temperature range up to 2300 K and atmospheric pressure of 0.1 MPa. Firstly, the composition of the feedstock was analyzed. As follows from Table 3, the main components of BW are oxygen (37.3 wt%), carbon (31.6 wt%) and water (20.8 wt%). There is a very large number of chemical reactions in the gasification process, but to assess the necessary amount of oxidants for complete carbon gasification (in ideal case), only the principal reaction can be taken into account:

Carbon partial oxidation: $C + \frac{1}{2}O_2 \rightarrow CO$ (ΔH = - 111 MJ/kmol).

Steam - carbon reaction: $C + H_2O \rightarrow CO + H_2$ (ΔH = +131 MJ/kmol).

Through the above chemical reactions it is possible to determine that only 72% by weight of both oxidants are consumed in the gasification. Thus, there is no need for an additional oxidant, justifying the choice for the plasma gasification process considering pyrolysis.

The variation in the composition of gases formed during the BW treatment process is shown in the figures (2) and (3) for temperatures from 500 K to 2300 K. In thermodynamic analysis were assumed the average values of the BW components according to Table 3.

As it can be seen in the Fig. 2 e 3, with an increase in the temperature the concentration of syngas (mixture of H_2 and CO) also increases up to the temperature around 1000 K. After this point, the syngas quantity remains constant until high temperatures when the process of dissociation of H_2 and CO begins. The produced gas is mainly composed of syngas which occupies 80.81% of the volume (36.98% CO and 43.83% H_2). The carbon is practically all consumed up to a temperature of 1000 K. At this temperature, in addition to syngas, there are major contaminants such as CH_4 methane (1.54%, decreasing above 1100 K), CO_2 carbon dioxide (9.37%, which decrease with temperature) and H_2O water vapor (7.74%, which increases with temperature). The nitrogen (N_2) concentration remains almost unchanged for temperatures above 1000 K and reaches 0.33 vol% only. Pollutant hydrogen sulfide (H_2S) is found at 0.13 - 0.10 vol% for temperature up to 1000 K and disappears after the temperature1800 K. The syngas chemical composition graphs in %vol as a function of temperature have results similar to those obtained by Messerle [9] for BT using air and steam as oxidant. The variations in the results occur due to the composition of the residues being a little different and also that for this work it is considered pyrolysis.

The Fig. 4 shows the variation of condensed matter in the simulation of the Brazilian BW plasma gasification process.

It is possible to observe through Fig. 4 that between the temperature of 300 and 373 K there is a marked decrease in the concentration of condensed matter, being 88%wt and 22%wt, respectively. Then there is a slower decrease until it disappears at a temperature of 1000 K. Thus, the plasma gasification process is ideal from 1000 K, where there is no presence of condensed matter.

Normally, the temperature inside the reactor varies due to presence of sources (plasma jet and/or electric arc) and sinks (reactor wall and exhaust channel). Assuming a 5% variation in α(T) to be allowed in the BW treatment process - can get a range of operating temperature, which to this work t can vary from 1000 K (limited by carbon consumption) to 1210 K. The Fig. 5 shows the variation of the syngas energy yield with temperature and the limits of variation energy yield are shown by dotted lines.

It can be seen in Fig. 5 that the highest yield can be achieved at a temperature of 1040 K, where $\alpha_{máx}$ is 2.25. For this temperature the syngas consists of 63.65 wt% (or 38.31 vol%) of carbon monoxide and 5.35 wt% (or 44.71 vol%) of hydrogen. For a temperature of 1040 K and its gas composition, the LHV of the gas is 13.47 MJ/kg.

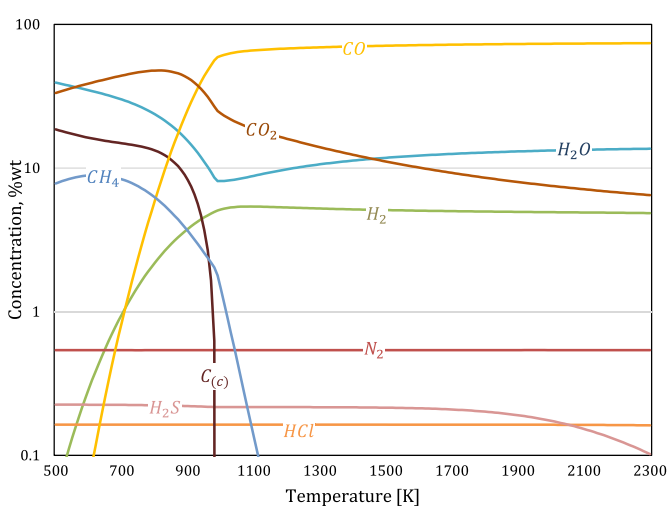

Fig. 2. Mass fraction, wt%, of species in produced gas of BW treatment.

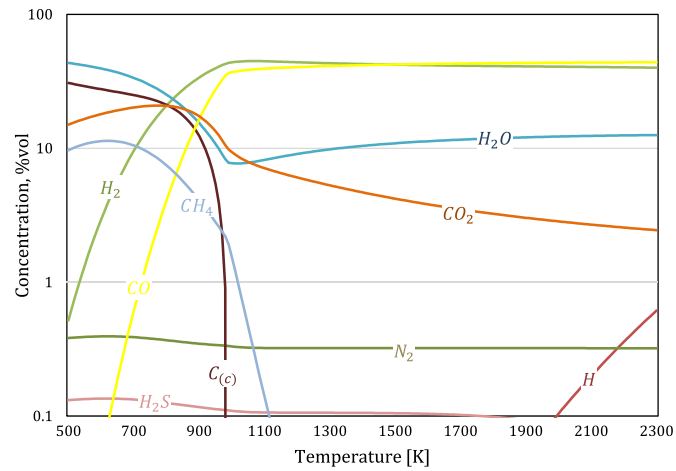

Fig. 3. Volume fraction, vol%, of species in produced gas of BW treatment.

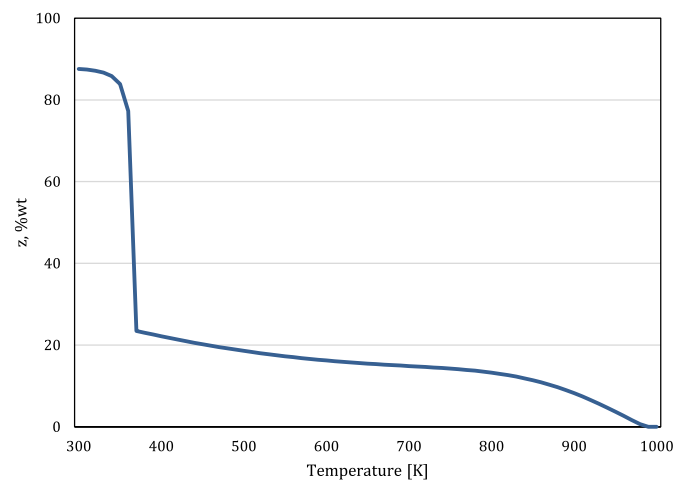

Fig. 4. Mass fraction of condensed matter.

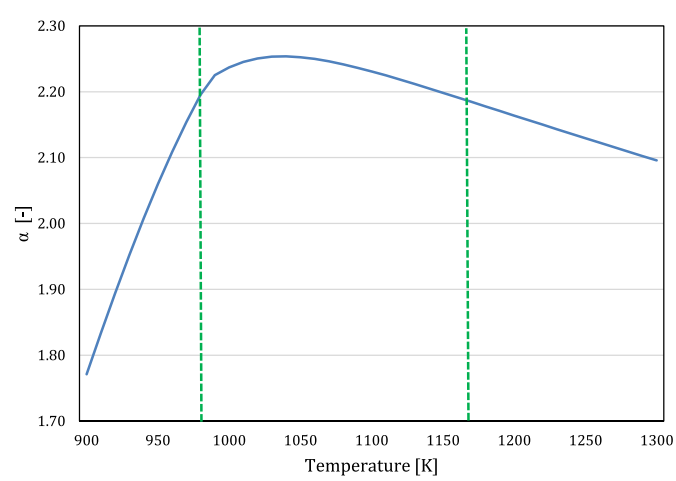

Fig. 5. Variation of the syngas energy yield with temperature.

Table 4
Composition in wt% and vol% of the elements of the gas produced in the plasma gasification of BW.

Element	H_2	H_2O	HCl	H_2S	N_2	CO	CO_2	CH_4
Composition, wt%	5.35	0.0829	0.16	0.22	0.54	63.65	21.21	0.57
Composition, vol%	44.7	7.75	0.07	0.11	0.32	38.30	8.12	0.59

Table 5
Comparison of syngas composition in % vol.

Results	Current Study	Erdogan [20]	Indrawan [21]	Messerle [9]
H_2	44.70	27.96	24.4–33.80	24.7 0–46.80
CO	36.98	20.77	22.00–30.90	28.7–38.10

The composition in wt% and vol% of the principal species are shown in Table 4.

Table 5 presents a comparison between the results of syngas composition in % vol of the Brazilian BW plasma gasification and other works available in the literature.

Comparing the results of this work with others found in the literature, it shows that the proportion of H2 and CO are similar, although the concentration is a little different, coming closer to the results obtained by Messerle [9]. This is due to the variation in the composition of the BW simulated by each author, so it is necessary to study the plasma gasification of Brazilian BW, to obtain the concentration of syngas according to the composition of biomedical waste produced in Brazil.

To estimate the potential for electricity production the calculations have been normalized to a BW flow rate of 1 kg/s, which results in the production of 0.93 kg/s of syngas and 0.07 kg/s of ashes.

The calculations show that consumption of electrical power in plasma torch needed for attain the working temperature (1040 K) is 6.3 MW for BW flow rate of 1 kg/s (1.75 kWh/kg). Masserle [9] obtained in his work a specific energy consumption of 1.60 and 1.70 kWh/kg for the processing of HW and BT, respectively, values similar to the ones in this work.

The Fig. 6 shows the electrical power produced in the internal combustion engine for electricity generation efficiency of 14% (ICE -14%) and 26% (ICE - 26%), for the gas turbine set with efficiency of 20% (GT - 20%) and 30% (GT - 30%) and the electrical power required in the plasma gasifier to process the BW (E_{torch}). The Fig. 7 shows the percentage that the internal combustion engine and gas turbine set is capable of supplying the electrical power in the plasma gasifier considering the minimum and maximum electricity generation efficiencies of each equipment.

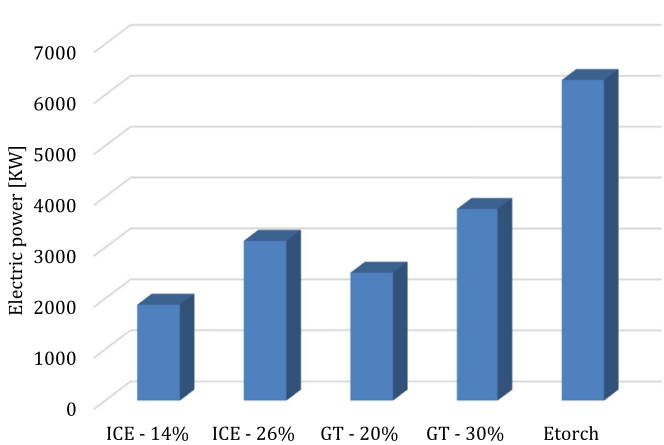

Fig. 6. Electric power production for different engines compared with the plasma torch consumption.

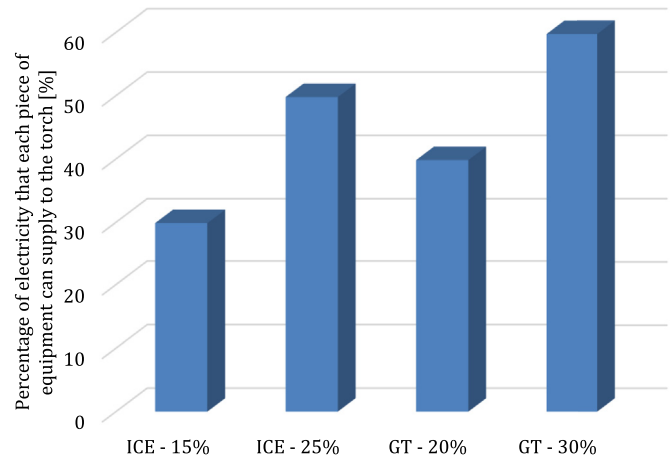

Fig. 7. Percentage of electricity that each piece of equipment can supply to the torch.

It can be seen from Figs. 6 and 7 that it is not possible to recover 100% of the electricity consumed by a plasma torch, which corresponds to 6292 kW. The internal combustion engine has electricity production potential of 1879–3132 kW for electricity generation efficiency of 14–26%, respectively, and thus can supply 29.87–49.78% of the electricity needed in the plasma torch, respectively. The gas turbine set, on the other hand, produces 2506–3758 kW of electrical power for electricity generation efficiency of 20–30%, respectively, supplying 39.82–59.74% of the energy required in the plasma torch, respectively. Thus, it is necessary that more studies are carried out in order to obtain ways to increase the efficiency of the plasma gasification process and also of the ICE and GTS to syngas so that this BW treatment system is more technically attractive. And economic studies are also needed to compare the plasma gasification process to incineration and autoclave, which have high costs and are the most used in Brazil.

4. Conclusion

Through thermodynamic analysis it is possible to verify that the best operating point of the plasma gasifier for the processing of Brazilian BW is when the gas energy yield is 2.25 at a temperature of 1040 K, for this temperature the composition of the syngas (%wt) is 63.65% CO and 5.35% H_2. The plasma torch has a high energy consumption, 6292 kW to process 1 kg/s of BW and produce 0.93 kg/s of syngas, and with the burning of this syngas in the ICE and GTS it is possible to produce up to 3132 kW and 3758 kW, respectively, so the capacity to supply the electrical power required in the torch is up to 49.78% and 59.74% with plasma gasification associated with ICE and GTS, respectively. Although plasma gasification technology requires a high consumption of electricity, it can become attractive from an economic point of view, in addition to being a technology that favors the environment and reduces risks to human health. And thermodynamic studies serve as a basis for economic studies in order to show the feasibility of implementing this process in Brazil and worldwide, in view of the high cost of disposing of BW by incineration, autoclave or microwave, and avoiding disposal inadequate BW, in view of the increased production of this type of waste due to the Covid-19 pandemic.

Credit author statement

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Visualization, Project administration. **Alexei Mikhailovich Essiptchouk**: Software, Formal analysis. **Lucas Pamplona Cardozo Costa**: Resources, Data curation. **José Luz Silveira**: Validation, Investigation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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